

Practice problems #23

1. True or False. Justify your answer:

(a) The set $B = \{1 + t^2, t + t^2, 1 + 2t + t^2\}$ in $\mathbf{P}_2$ is linearly independent.

(b) The set $B = \left\{ \begin{bmatrix} 1 & 2 & -5 & 11 & -3 \\ 2 & 4 & -5 & 15 & 2 \\ 1 & 2 & 0 & 4 & 5 \\ 3 & 6 & -5 & 19 & -2 \end{bmatrix}, \begin{bmatrix} 0 & 2 & -5 & 11 & -3 \\ 2 & 4 & -5 & 15 & 2 \\ 1 & 2 & 0 & 4 & 5 \\ 3 & 6 & -5 & 19 & -2 \end{bmatrix} \right\}$ in $\mathbf{R}^{4 \times 5}$ is linearly independent.

2. Find a basis for the set of vectors (x, y, z) in $\mathbf{R}^3$ such that $x + 2y + z = 0$.

3. Find bases for $\text{Nul } A$ and $\text{Col } A$, where A is row equivalent to B :

$$A = \begin{bmatrix} 1 & 2 & -5 & 11 & -3 \\ 2 & 4 & -5 & 15 & 2 \\ 1 & 2 & 0 & 4 & 5 \\ 3 & 6 & -5 & 19 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 0 & 4 & 5 \\ 0 & 0 & 5 & -7 & 8 \\ 0 & 0 & 0 & 0 & -9 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

4. Let $v_1 = \begin{bmatrix} 4 \\ -3 \\ 7 \end{bmatrix}$, $v_2 = \begin{bmatrix} 1 \\ 9 \\ -2 \end{bmatrix}$, $v_3 = \begin{bmatrix} 7 \\ 11 \\ 6 \end{bmatrix}$, and $H = \text{Span}\{v_1, v_2, v_3\}$. It can be verified that $4v_1 + 5v_2 - 3v_3 = 0$. Use this information to find a basis for H .

5. True or False. Justify:

(a) Suppose $v_1, \dots, v_4$ are vectors in $\mathbf{R}^4$ such that $\mathbf{R}^4 = \text{Span}\{v_1, \dots, v_4\}$. Then $\{v_1, \dots, v_4\}$ is a basis for $\mathbf{R}^4$.

(b) Let $B = \{v_1, \dots, v_n\}$ be a linearly independent set in $\mathbf{R}^n$. Then B is a basis for $\mathbf{R}^n$.

(c) Every vector space has a basis.

Answers are on the back.

Answers:

#1(a): True. Consider the equation $c_1(1 + t^2) + c_2(t + t^2) + c_3(1 + 2t + t^2) = 0$.

#1(b): True. Justify!

#2: This problem is essentially asking to find a basis for $\text{Nul } A$ for an appropriate matrix

A. One possible answer: $\left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$

#3: Basis for $\text{Nul } A$: $\left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -4 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right\}$

Basis for $\text{Col } A$: $\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} -5 \\ -5 \\ 0 \\ -5 \end{bmatrix}, \begin{bmatrix} -3 \\ 2 \\ 5 \\ -2 \end{bmatrix} \right\}$

#4: $\{v_1, v_2\}$ or $\{v_1, v_3\}$ or $\{v_2, v_3\}$ are all possible answers.

#5(a): True. Let $A = [v_1 \cdots v_4]$. Then, A is a 4×4 matrix. If $\mathbf{R}^4 = \text{Span } \{v_1, \dots, v_4\}$, what can you say about the columns of A ?

#5(b): True. Note that the matrix $A = [v_1 \cdots v_n]$ is $n \times n$.

#5(c): False. Give an example of a vector space with no basis.